

WHALE SHARK



Whale Shark Final Presentation

Madelyne Zhang · Leo Lyu

Supervisor: Vianey Leos Barajas

University of Toronto

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Original Dataset and Data Processing

Original Dataset

- Irregular shark tracking data
- Observations recorded at uneven time intervals
- Includes longitude, latitude, measurement error type and timestamps

Data Processing

- Divided each shark track into separate blocks
- Removed segments if consecutive observations were more than 7 days
- Retain blocks with at least 20 observations
- Ensured consistent and reliable movement sequence for model fitting

Fit SSM and MPM to get move persistence

State-Space Model (SSM)

- Fitted a state-space model to account for:
 - observation error
 - irregular sampling intervals
- Estimated the underlying movement path
- Regularized tracks to 24-hour time steps

Movement Persistence Model (MPM)

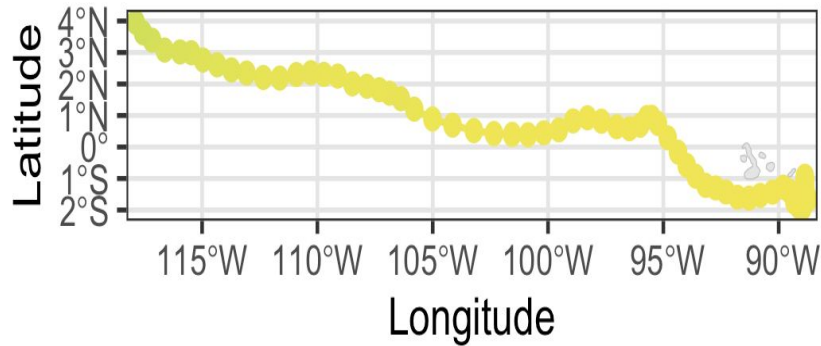
- Applied a movement persistence model to SSM outputs
- Estimated movement persistence parameter (γ)
- γ ranges from 0 to 1
- Describes the degree of directional movement

$$v_t = \gamma_t v_{t-1} + \eta_t, \quad \gamma_t \in (0, 1)$$

Low move persistence vs High move persistence Track

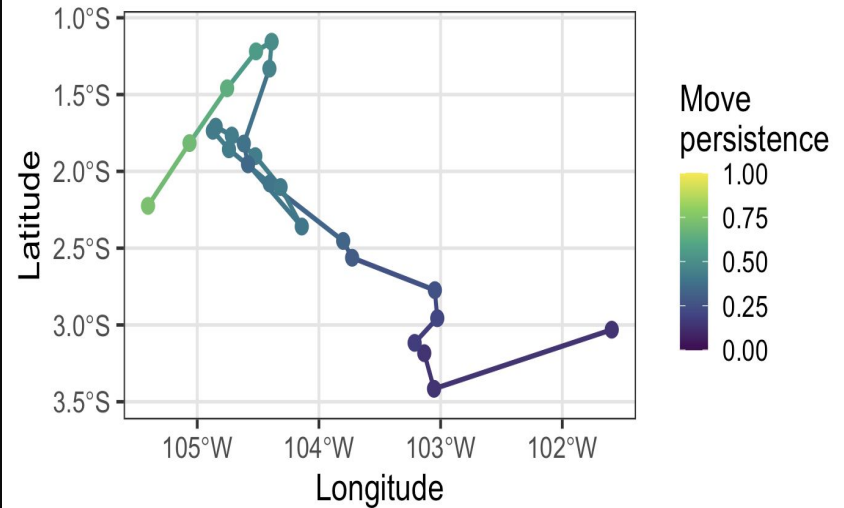
High move persistence shark track

151677_2 mean g = 0.973

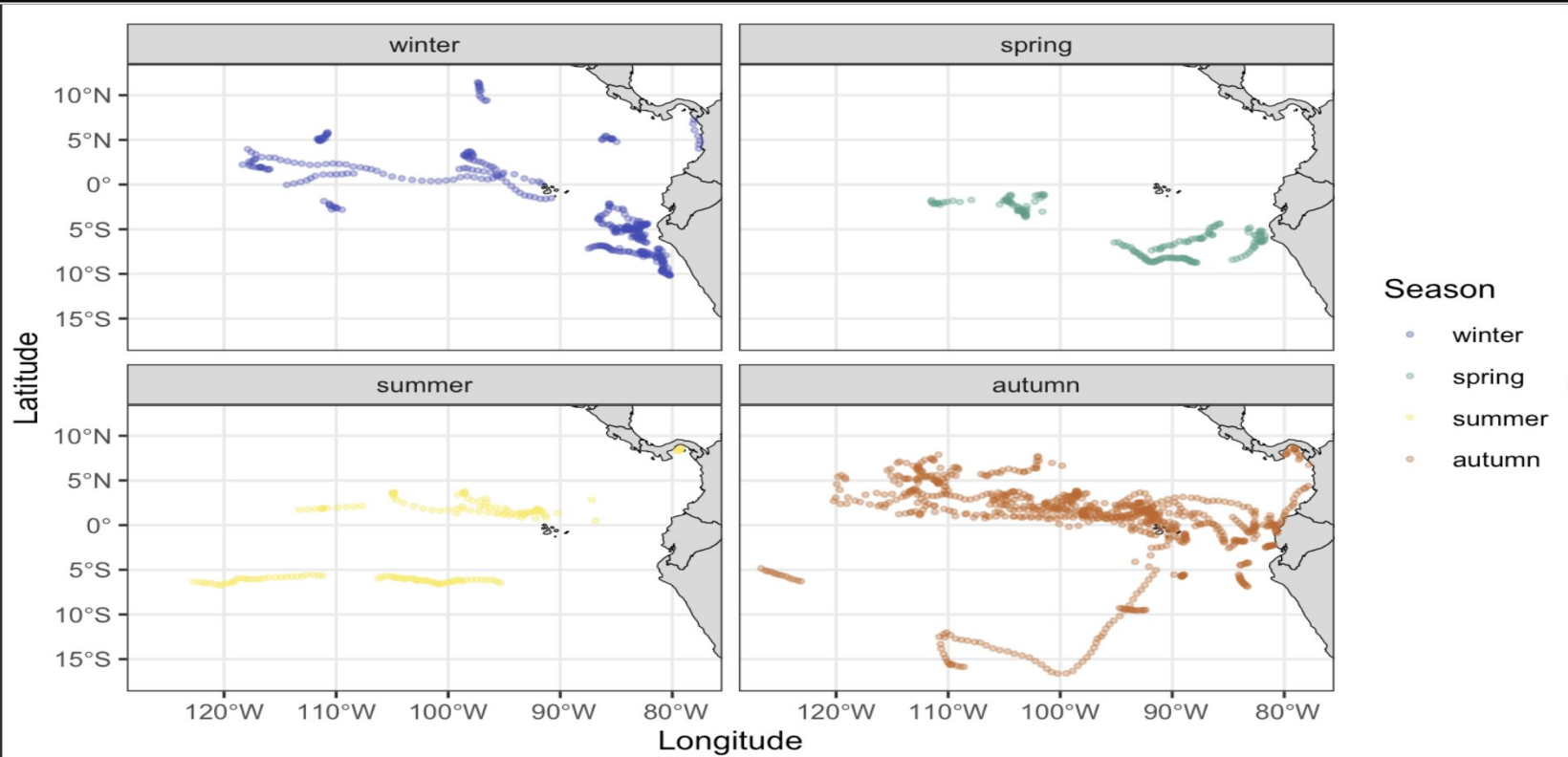


Low move persistence shark track

184027_8 mean g = 0.389



Move persistence by Season



Extract environmental covariates for Modeling

- Used predicted latitude and longitude and predicted daytime from the SSM-fitted tracks
- Extracted environmental covariates at each predicted location
- Covariates included:
 - Sea Surface Temperature (SST)
 - Bathymetry

Variable description

- SST: temperature of the ocean surface
- Bathymetry: ocean depth or underwater terrain

Results

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Environmental Terms

Sea Surface Temperature

Linear

Linear term for sea surface temperature

Season

Fixed Effect

Fixed effect with four seasonal levels

Distance to Coast * Coast Type

Interaction

Interaction term allowing the coastal effect to differ between island and continent settings

Spatial Location

Smooth

2D spatial smooth $s(\text{lat}, \text{lon})$ capturing spatial structure

Bathymetry

Linear

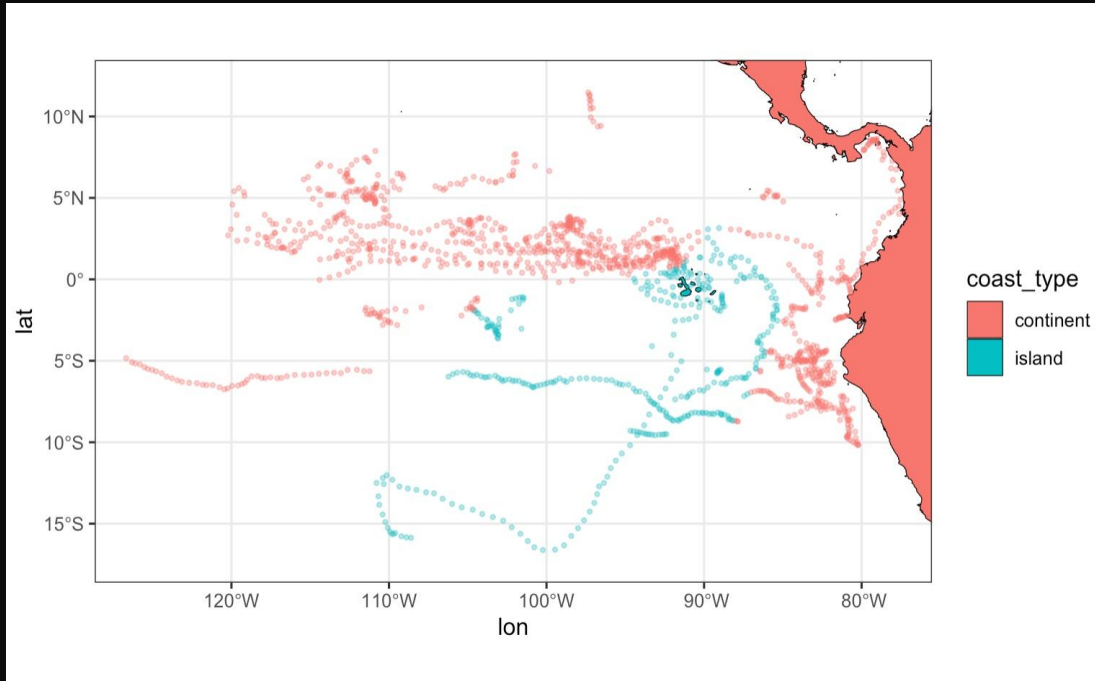
Seafloor depth; included as a linear term after excluding land observations

Individual ID

Random Effect

Random effect $s(\text{id})$ for repeated observations within shark

Shark Locations Classified by Coast Type



Coast Type Definitions

- **continent**: Location nearest to mainland coast
- **island**: Location nearest to oceanic islands near Ecuador

Distance to Coast

Standardized distance to the nearest coastline, modeled in interaction with coast type.

Motivation

Whale sharks may respond differently to continental and island coastlines, suggesting different habitat use patterns.

Model Output

Final GAMM after standardizing continuous predictors

0.798

Adjusted R^2

86%

Deviance Explained

1,509

Observations Used

Family: Beta regression(16.003)

Link function: logit

Formula:

$\text{gamma_beta} \sim \text{sst_z} + \text{bathymetry_z} + \text{dist_to_coast_km_z} * \text{coast_type} + \text{season} + \text{s}(\text{lat}, \text{lon}, \text{bs} = "tp", k = 50) + \text{s}(\text{id}, \text{bs} = "re")$

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.32531	0.24107	5.498	3.85e-08	***
sst_z	0.15207	0.03881	3.919	8.90e-05	***
bathymetry_z	0.10834	0.03564	3.039	0.002371	**
dist_to_coast_km_z	1.68870	0.39179	4.310	1.63e-05	***
coast_typeisland	0.10546	0.11237	0.938	0.348009	
seasonspring	-0.17768	0.14961	-1.188	0.234994	
seasonsummer	-0.62971	0.14840	-4.243	2.20e-05	***
seasonautumn	0.19240	0.11305	1.702	0.088762	.
dist_to_coast_km_z:coast_typeisland	-0.47992	0.12835	-3.739	0.000185	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value	
s(lat,lon)	44.93	47.98	1079	<2e-16	***
s(id)	49.85	52.00	2406	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.798 Deviance explained = 86%

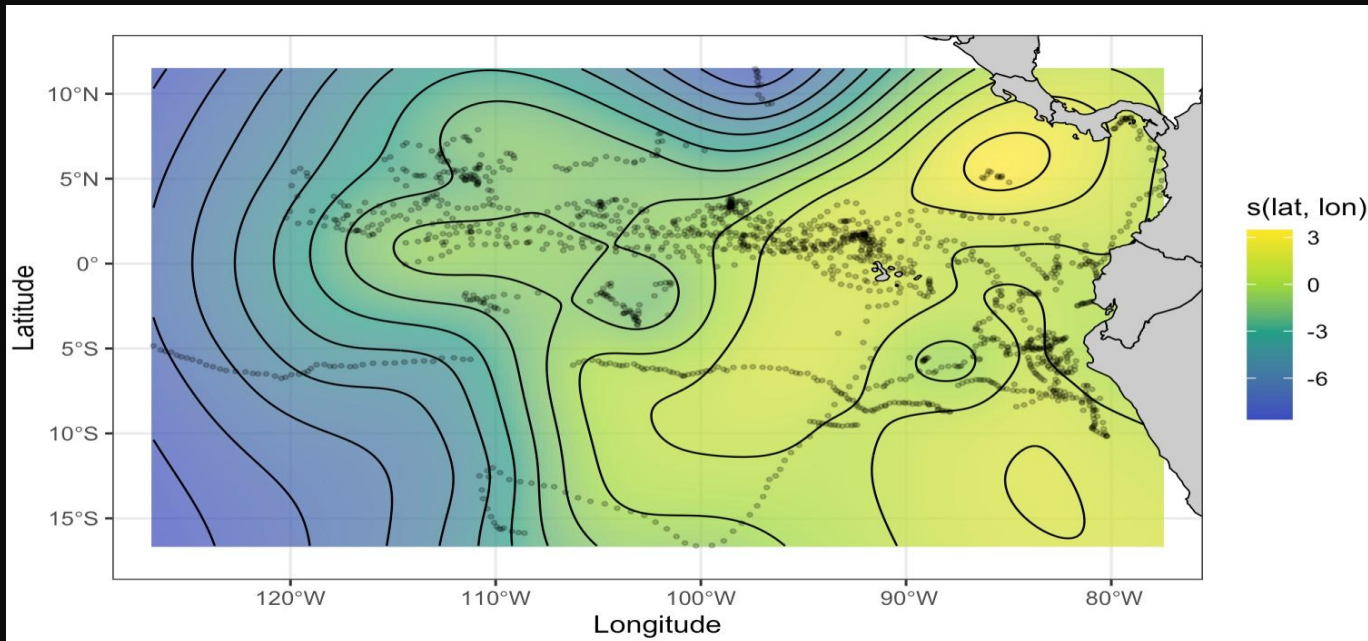
-REML = -1505 Scale est. = 1 n = 1509

Model Output

Final GAMM after standardizing continuous predictors

SST	Higher SST was associated with higher movement persistence
Bathymetry	Shallower bathymetry was associated with higher movement persistence
Distance to coast	Positive effect in continent-associated settings
Island coast type	No significant main effect
Summer	Lower movement persistence than winter
Distance to coast * island	Coastal-distance effect was weaker in island settings

Spatial Smooth $s(\text{lat}, \text{lon})$



The spatial effect was clearly non-linear across the study region.

Higher fitted values ($\gamma_{\square}$) were concentrated in the eastern study region.

Substantial residual spatial structure remained after adjusting for environmental covariates.

Main Takeaways

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Conclusions

- Overall model fit was strong, with 86% deviance explained.
- Shallower waters and higher SST were associated with higher movement persistence.
- The distance-to-coast effect differed between island and continent settings.
- Summer was associated with lower movement persistence relative to winter.
- Spatial and between-individual heterogeneity remained important after adjustment.